

Automated UI Testing for Headlines

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Project Overview

● THE APP

Headlines is a MetroNational Teams app that::

- Creates AI generated headline feeds for tenants
- A tenant profiles sections that clearly and easily lays out tenant and lease information

● THE ASSIGNMENT

Quality Assurance testing done through the design and implementation of Playwright test scripts.

This is done to ensure that the app's user interface is working properly in a frequent and efficient manner.

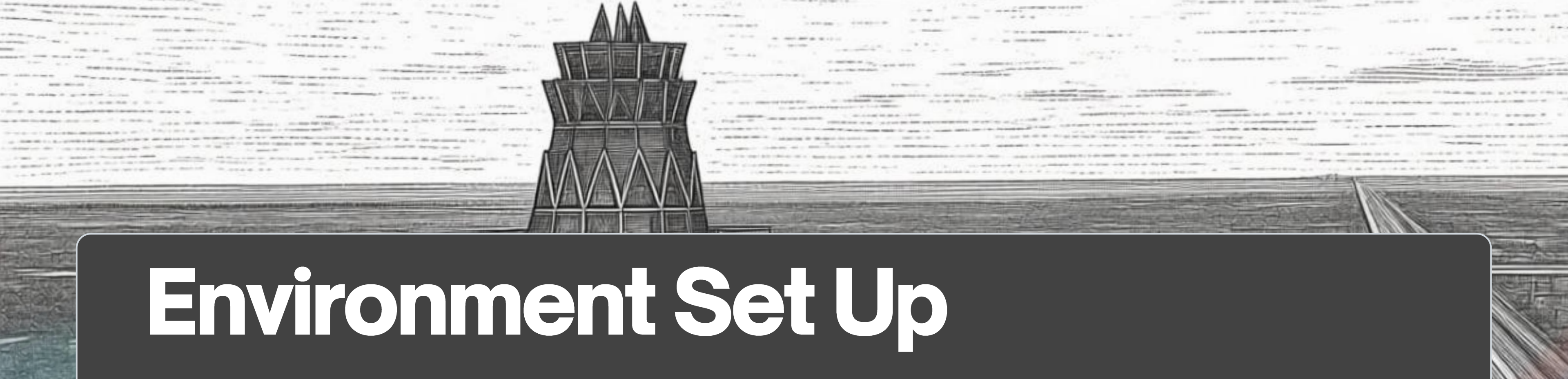
● THE GOAL

Create automated playwright tests that will run on a nightly basis and report back test results.

Eliminates the need for manual testing and provides developers regular reports of what features need their attention.

Timeline

- 01** Environment Set Up
- 02** Manual QA
- 03** Headlines Feed Playwright Test
- 04** Reactions Playwright Test
- 05** Tenant Profile Playwright Test
- 06** Notification Toggle Playwright Test
- 07** Automation



Environment Set Up

Task:

- Need to set up Playwright
- Set up an effective and efficient method to navigate past the login authentication stage automatically

Skills Utilized:

- JavaScript programming
- Algorithms and logic
- Debugging techniques
- Problem-solving strategies

Obstacles/Blockers:

- New to Playwright
- Still familiarizing myself with the Headlines app UI
- Navigating security measures and guidelines

Solution:

- Researched and discussed with team members on the most secure paths to use
- I landed upon using a storage state that automatically began all future tests past the login page

Manual QA

01

Quality Assurance

Identify bugs, verify features, and ensure the application provides a reliable and user-friendly experience before it's released.

02

Process

1. Navigate through the application
2. Verify functionality of every feature
3. Observe expected behavior
4. Record findings: Document issues if discovered

03

Insights

1. Provided a fresh set of eyes to “judge” the app
2. Understood how users interact with the application
3. Identified repetitive tasks suitable for automation
4. Designed more accurate automated tests

04

Feature Catalog

Created an in-depth document, detailing:

- Each feature of the app
- How to access and operate
- Any known limitations

Headlines

Headlines feed loads and displays current headlines – Pass

Each headline shows expected fields – Pass

Reactions apply and persist on refresh – Pass

Comments apply and are visible on return – Pass

Filtering by category shows only that type – Pass

Historical headlines accessible – Pass

Notification toggle works – Pass

1. Tag Formatting: Some tags do not include spaces between words (“LeaseExpiration”, “PersonalInjury”, “CalendarEvents”).

2. Typographical Errors: Small typo identified in HR headline dated 6/10: “Officer” is misspelled as “officr”.

3. The layout of the post new headline page is slightly misaligned and would look better if centered or more right-aligned.

4. Headline Description Formatting: Some inconsistency in the usage of commas, as observed in recent security headlines.

5. Date Formatting: Inconsistent date formatting across headline descriptions (Rocks section headlines go from “May 11 2026” to “2026-05-18”).

Standardization would help in making a more seamless look.

6. The total number of headlines displayed in the filtering tab buttons is unclear in purpose. It may be more meaningful to display the number of today’s headlines, which could also help draw attention to newly added content.

Tenant Profiles

Link-through to Tenant Profile opens the correct profile – Pass

Profile tab loads without a blank screen – Pass

All expected fields present and populated correctly – Pass

Multi-lease dropdown updates fields on switch – Pass

AI insights render with content – Pass

Hamburger menu navigates through app correctly - Pass

1. Murphy Exploration & Production

a. Two separate tenant profiles, one is multi lease while the other is a singular lease, should be merged into one profile.

b. Both profiles are missing the spaces around the ‘&’, and one profile appears to have its final character cut off.

2. Some profiles, like AECOM, uses a separate lease-selection page prior to displaying lease information which is inconsistent.

3. Duplicate lease profile for AECOM Technical Services, matches Two Memorial City Lease in AECOM’s main profile

3. Briefing Page Alignment: The page appears slightly misaligned compared to the rest of the pages within the profiles.

4. Parking Information

a. Blank values for the number of parking spots, particularly for the number of unreserved spots

b. The purpose of the "Show Original" option on the Parking Information page is unclear.

Headlines Feed Tests



Feed Validation

- Ensure the headlines page within the app is navigated to
- Check for today's headline, if none are present then check for the fallback message
- Navigate through each tab and check for the headline/fallback message

Headline Contents

- If there is a headline today, the test will check for a title, category, summary, and date
 - ◆ Category: "WORK ORDERS"
 - ◆ Date: "Today"
 - ◆ Title: "Suite Remediation Pending Tenant Sign-Off"
 - ◆ Summary: "Hartford Steam Boiler Ins leads unresolved WOs at \$37.2K. 2 orders pending e-sign: suite clean and paint at 820 Gessner Rd."

Historical Headlines

- Clicks on the "View Older Headlines" button if present
- Checks that upon click, historical headlines do in fact appear as intended

User Reaction Tests

Liking

- Likes the most recent headline available
- Ensures the like count adjusts accordingly
- Refreshes, and checks that the like persisted
- Repeats the process again to test the unlike ability

Saving

- Saves the most recent headline available
- Navigates to the saved headlines page to ensure it has correctly appeared there
- Unsaves that very headline
- Checks that the headline is no longer appearing on saved page.

Commenting

- Checks the active chats page for threads
 - In the case of no active chats, the test will check for the correct fallback message to be present
- Expands each one to ensure comments are present and visible to users

Tenant Profile Tests

Profile Search

1. Navigate to tenant profile page
2. Type in the hard coded tenant names
3. Select their correct profile

Lease Details

1. Capture the tenant name, square footage, lease date, lease ID, building, floor, and suite from the Lease tab
2. Match these details to those within the tenant variable, and report any mismatches

Further Info

1. Parking tab: Check for parking info box formatting or fallback message
2. News tab: Checks for AI insight headlines or fallback message
3. Contacts tab: Checks for valid contact information or fallback message

Multi-Lease

1. Check if the tenant profile is multi-lease
2. If so, run the previous tests on the default lease
3. Navigate back into the same profile, and switch to the next lease
4. Repeat the tests and all leases within the profile

Notification Toggle Test

- Navigate to the Notifications page
- Locate the main notification toggle button
- Click the button to effectively turn notifications off
- Refresh app and navigate back, confirm that the notifications have remained off
- Click the notification toggle button again to turn notifications on
- Refresh app and navigate back, confirm that the notifications are still on

Main Suite Runner

Using a main suite runner, all tests are run consecutively, and a final summary report is created

```
=====
                SUITE RUN SUMMARY
=====
Tabs Filter Test:      PASSED
Past Headlines Test:   PASSED
Comments Visibility Test: PASSED
Reactions Test:       PASSED
Saving Test:          PASSED
Tenant Profile Test:   PASSED
Notification Toggle Test: PASSED
=====
```

This main suite will be automated to run on a nightly basis, and these summary results will be sent to developers Teams chat.

Reflection

01

Business Value

Delivered automated and reusable Playwright UI test scripts. Reduces repetitive manual testing, improved testing consistency, and support for future UI testing efforts

02

Technical Growth

- Learned a modern automation framework
- Applied programming concepts to a real project
- Improved debugging skills
- Better understanding of software testing

03

Professional Growth

- Worked through real-world technical challenges
- Improved confidence solving problems
- Experienced how software supports business objectives first-hand
- Communicated and tracked updates with my team

04

Future Growth

- Increase overall test coverage and reporting with more error message catching
- Utilize newfound knowledge and experience to continue exploring QA and product management more

**Thank
You**